

Part I

Basic Properties of the Natural Logarithm:

Summary

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Basic properties of the natural logarithm:

- $\ln(yz) = \ln(y) + \ln(z)$
- $\ln\left(\frac{y}{z}\right) = \ln(y) - \ln(z)$
- $\ln(y^z) = z \ln(y)$
- For any function g : $\ln(e^{g(x)}) = g(x)$
- $\frac{d}{dx}(\ln(f(x))) = \frac{f'(x)}{f(x)}$.

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 (a) Write as an exponential with base 5. 7^{3x} .

Explanation. $7 = 5^{\log_5(7)}$, so

$$7^{3x} = \left(5^{\log_5(7)}\right)^{3x} = 5^{3x \log_5(7)}$$

(b) Write in terms of the natural logarithm. $\log_3(4)$.

Explanation. The change of base formula says $\log_a(u) = \frac{\log_b(u)}{\log_b(a)}$, so

$$\log_3(4) = \frac{\ln(4)}{\ln(3)}$$

(c) Expand the following: $\log_{1/2} \left(\frac{6x^5(2 + \tan(x))^x}{\sqrt[5]{e^{4x} + 1}} \right)$.

Explanation.

$$\log_{1/2} \left(\frac{6x^5(2 + \tan(x))^x}{\sqrt[5]{e^{4x} + 1}} \right) = \log_{1/2}(6) + 5 \log_{1/2}(x) + x \log_{1/2}(2 + \tan(x)) - \frac{1}{5} \log_{1/2}(e^{4x} + 1)$$

Problem 2

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Problem 2 Find all real numbers x which satisfy each of the following equations.

(a) $\log_x(25) = 2$.

Explanation. Recall that, by definition of the inverse to an exponential function,

$$\log_b(x) = y \iff x = b^y.$$

Using this relationship we have $\log_x(25) = 2 \iff x^2 = 25$. Therefore

$$\begin{aligned} x^2 = 25 &\implies x = \pm 5, \\ &\implies x = 5 \quad (\text{base of log is always } > 0). \end{aligned}$$

Therefore $x = 5$ is the only solution to $\log_x(25) = 2$.

(b) $7^x = 15$

Explanation. Similar to the previous problem,

$$\log_b(x) = y \iff x = b^y.$$

Using this relationship we have $7^x = 15 \iff x = \log_7(15)$. Therefore $x = \log_7(15)$ is the only solution to $7^x = 15$.

(c) $\ln(x) + 1 = 0$.

Explanation. Similar to the previous two problems we'll use the relationship

$$\ln(x) = y \iff x = e^y.$$

Before applying this relationship we perform a bit of algebra first:

$$\ln(x) + 1 = 0 \implies \ln(x) = -1.$$

Now we have $\ln(x) = -1 \iff x = e^{-1}$. Therefore $x = e^{-1} (= 1/e)$ is the only solution to $\ln(x) + 1 = 0$.

Problem 3

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Problem 3 True or False:

(1) If $f(x) = (x - 2)^x$, then $f'(x) = x(x - 2)^{x-1}$.

Explanation. False. Any time that you have a function of x raised to a function of x , in order to compute the derivative you need to use logarithmic differentiation (or something equivalent).

Correct derivative of f :

$$\begin{aligned} f(x) = (x - 2)^x &\implies f(x) = e^{x \ln(x-2)} \\ \implies f'(x) &= e^{x \ln(x-2)} \cdot \left(1 \cdot \ln(x-2) + x \cdot \frac{1}{x-2} \right) \\ \implies f'(x) &= (x - 2)^x \cdot \left(\ln(x-2) + \frac{x}{x-2} \right) \end{aligned}$$

(2) If $f(x) = (3x)^x$, then $f'(x) = (3x)^x \ln(3x)$.

Explanation. False. Same as part (1).

Correct derivative of f :

$$\begin{aligned} f(x) = (3x)^x &\implies f(x) = e^{x \cdot \ln(3x)} \\ \implies f'(x) &= e^{x \cdot \ln(3x)} \left(1 \cdot \ln(3x) + x \cdot \frac{3}{3x} \right) \\ \implies f'(x) &= (3x)^x (\ln(3x) + 1) \end{aligned}$$

Problem 4

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Problem 4 Find the derivatives of the following functions:

(a) $f(x) = x^{e^x} + 7x$

Explanation. $f'(x) = \frac{d}{dx} (x^{e^x}) + \frac{d}{dx} (7x) = \frac{d}{dx} (x^{e^x}) + 7$. So the real problem is to find $\frac{d}{dx} (x^{e^x})$.

We will use logarithmic differentiation.

First, we take natural logarithm of both sides of the equation $g(x) = x^{e^x}$.
 $\ln(g(x)) = e^x \ln(x)$.

Now, we differentiate both sides.

$$\frac{g'(x)}{g(x)} = e^x \ln(x) + \frac{e^x}{x}.$$

Next, we solve for $g'(x)$.

$$g'(x) = g(x) \left(e^x \ln(x) + \frac{e^x}{x} \right).$$

Now, we substitute.

$$g'(x) = x^{e^x} \left(e^x \ln(x) + \frac{e^x}{x} \right).$$

ALTERNATIVE APPROACH

$$\begin{aligned} \frac{d}{dx} (x^{e^x}) &= \frac{d}{dx} (e^{\ln(x)e^x}) \\ &= \frac{d}{dx} (e^{e^x \ln(x)}) \\ &= e^{e^x \ln(x)} \left(e^x \ln(x) + \frac{e^x}{x} \right) \\ &= x^{e^x} \left(e^x \ln(x) + \frac{e^x}{x} \right). \end{aligned}$$

$$\text{Thus, } f'(x) = x^{e^x} \left(e^x \ln(x) + \frac{e^x}{x} \right) + 7.$$

(b) $g(x) = (\ln(x) + 9)^{\sec(x^4)}$

Explanation. We will use logarithmic differentiation.

First, we take natural logarithm .

Problem 4

$$\ln(g(x)) = \sec(x^4) \ln(\ln(x) + 9)$$

Then, we differentiate.

$$\frac{g'(x)}{g(x)} = \sec(x^4) \tan(x^4) 4x^3 \ln(\ln(x) + 9) + \frac{\sec(x^4)}{x(\ln(x) + 9)}$$

We multiply by $g(x)$.

$$g'(x) = g(x) \left(4x^3 \sec(x^4) \tan(x^4) \ln(\ln(x) + 9) + \frac{\sec(x^4)}{x(\ln(x) + 9)} \right)$$

We substitute.

$$g'(x) = (\ln(x) + 9)^{\sec(x^4)} \left(4x^3 \sec(x^4) \tan(x^4) \ln(\ln(x) + 9) + \frac{\sec(x^4)}{x(\ln(x) + 9)} \right)$$

ALTERNATIVE APPROACH

$$\begin{aligned} g'(x) &= \frac{d}{dx} \left((\ln(x) + 9)^{\sec(x^4)} \right) \\ &= \frac{d}{dx} \left(e^{\sec(x^4) \ln(\ln(x) + 9)} \right) \\ &= e^{\sec(x^4) \ln(\ln(x) + 9)} \left(4x^3 \sec(x^4) \tan(x^4) \ln(\ln(x) + 9) + \sec(x^4) \frac{\frac{1}{x}}{\ln(x) + 9} \right) \\ &= (\ln(x) + 9)^{\sec(x^4)} \left(4x^3 \sec(x^4) \tan(x^4) \ln(\ln(x) + 9) + \frac{\sec(x^4)}{x(\ln(x) + 9)} \right). \end{aligned}$$

$$(c) \ f(x) = \frac{(x+1)^5 (\sin(x) + 5)^4}{(x^2 + 5) \sqrt{x-3}}$$

Explanation.

$$\begin{aligned} \ln(f(x)) &= \ln(x+1)^5 + \ln(\sin(x) + 5)^4 - \ln(x^2 + 5) - \ln \sqrt{x-3} \\ &= 5 \ln(x+1) + 4 \ln(\sin(x) + 5) - \ln(x^2 + 5) - \frac{1}{2} \ln(x-3) \end{aligned}$$

Now we differentiate.

$$\frac{f'(x)}{f(x)} = \frac{5}{x+1} + \frac{4 \cos(x)}{\sin(x) + 5} - \frac{2x}{x^2 + 5} - \frac{1}{2(x-3)}$$

Then we solve for $f'(x)$.

$$f'(x) = f(x) \left(\frac{5}{x+1} + \frac{4 \cos(x)}{\sin(x)+5} - \frac{2x}{x^2+5} - \frac{1}{2(x-3)} \right) \text{ Then we substitute.}$$

$$f'(x) = \frac{(x+1)^5 (\sin(x)+5)^4}{(x^2+5)\sqrt{x-3}} \left(\frac{5}{x+1} + \frac{4 \cos(x)}{\sin(x)+5} - \frac{2x}{x^2+5} - \frac{1}{2(x-3)} \right)$$

$$(d) \ h(x) = \frac{(x^2-7)^5}{\cos^7(x^2-5)}$$

Explanation. Since $h(x)$ changes sign on its domain, we write :

$$\begin{aligned} \ln |h(x)| &= \ln \left(\frac{|x^2-7|^5}{|\cos^7(x^2-5)|} \right) \\ &= 5 \cdot \ln(|x^2-7|) - 7 \cdot \ln(|\cos(x^2-5)|) \end{aligned}$$

Differentiate both sides with respect to x , (explanation given on the next page):

$$\frac{h'(x)}{h(x)} = 5 \frac{2x}{x^2-7} - 7 \frac{-\sin(x^2-5)2x}{\cos(x^2-5)}$$

$$\begin{aligned} h'(x) &= h(x) \left(5 \frac{2x}{x^2-7} - 7 \frac{-\sin(x^2-5)2x}{\cos(x^2-5)} \right) = \\ &= \frac{(x^2-7)^5}{\cos^7(x^2-5)} \left(\frac{10x}{x^2-7} + \frac{14x \sin(x^2-5)}{\cos(x^2-5)} \right) = \\ &= \frac{(x^2-7)^4}{\cos^7(x^2-5)} (10x + 14x(x^2-7) \tan(x^2-5)) \end{aligned}$$

EXPLANATION: The function $h(x)$ assumes negative values on some intervals. On those intervals

$$(\ln(|h(x)|))' = (\ln(-h(x)))' = \frac{-h'(x)}{-h(x)} = \frac{h'(x)}{h(x)},$$

Also, e.g., if $x^2-7 < 0$

$$(\ln(|x^2-7|))' = (\ln(-(x^2-7)))' = \frac{-2x}{-(x^2-7)} = \frac{2x}{x^2-7}.$$